

shaped surface shown in Figure 12.25. The cross-sections with x fixed are horizontal lines obtained by cutting the surface by a plane perpendicular to the x -axis. This surface is called a *parabolic cylinder*, because it is formed from a parabola in the same way that an ordinary cylinder is formed from a circle; it has a parabolic cross-section instead of a circular one.

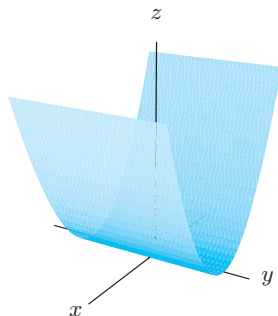


Figure 12.25: A parabolic cylinder $z = x^2$

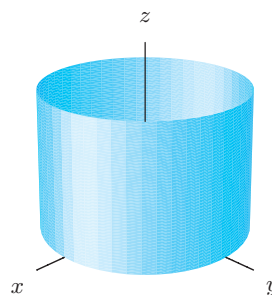


Figure 12.26: Circular cylinder $x^2 + y^2 = 1$

Example 5 Graph the equation $x^2 + y^2 = 1$ in 3-space.

Solution Although the equation $x^2 + y^2 = 1$ does not represent a function, the surface representing it can be graphed by the method used for $z = x^2$. The graph of $x^2 + y^2 = 1$ in the xy -plane is a circle. Since z does not appear in the equation, the intersection of the surface with any horizontal plane will be the same circle $x^2 + y^2 = 1$. Thus, the surface is the cylinder shown in Figure 12.26.

Exercises and Problems for Section 12.2

Exercises

1. Figure 12.27 shows the graph of $z = f(x, y)$.

- Suppose y is fixed and positive. Does z increase or decrease as x increases? Graph z against x .
- Suppose x is fixed and positive. Does z increase or decrease as y increases? Graph z against y .

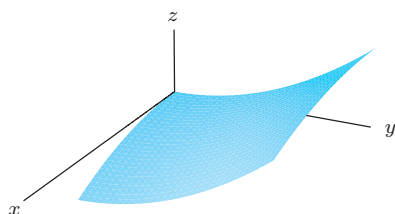


Figure 12.27

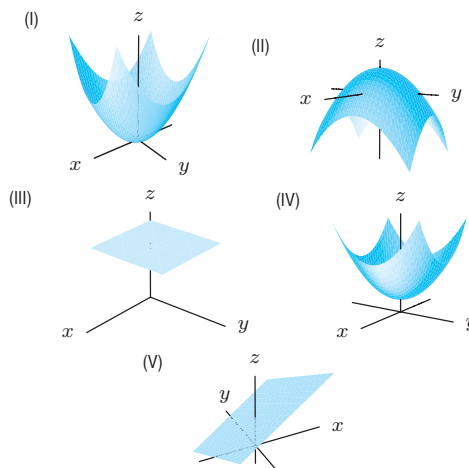


Figure 12.28

2. Without a calculator or computer, match the functions with their graphs in Figure 12.28.

- | | |
|-------------------------|-------------------------|
| (a) $z = 2 + x^2 + y^2$ | (b) $z = 2 - x^2 - y^2$ |
| (c) $z = 2(x^2 + y^2)$ | (d) $z = 2 + 2x - y$ |
| (e) $z = 2$ | |

3. Without a calculator or computer, match the functions with their graphs in Figure 12.29.

- | | |
|-------------------------------|---------------------------|
| (a) $z = \frac{1}{x^2 + y^2}$ | (b) $z = -e^{-x^2 - y^2}$ |
| (c) $z = x + 2y + 3$ | (d) $z = -y^2$ |
| (e) $z = x^3 - \sin y$ | |

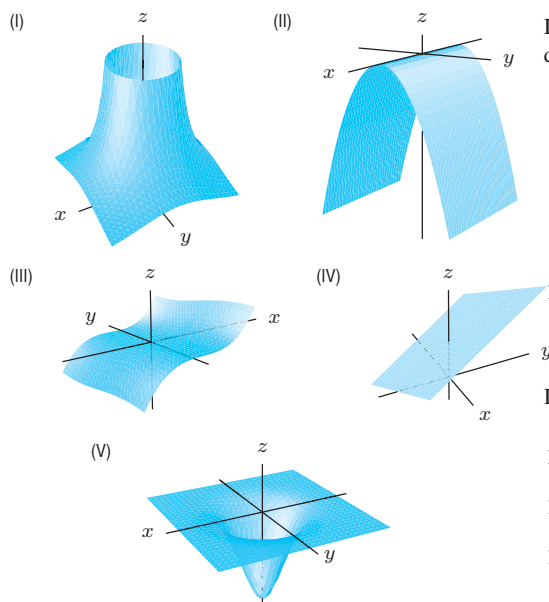


Figure 12.29

In Exercises 4–11, sketch a graph of the surface and briefly describe it in words.

4. $z = 3$
5. $x^2 + y^2 + z^2 = 9$
6. $z = x^2 + y^2 + 4$
7. $z = 5 - x^2 - y^2$
8. $z = y^2$
9. $2x + 4y + 3z = 12$
10. $x^2 + y^2 = 4$
11. $x^2 + z^2 = 4$

In Exercises 12–14, find the equation of the surface.

12. A cylinder of radius $\sqrt{7}$ with its axis along the y -axis.
13. A sphere of radius 3 centered at $(0, \sqrt{7}, 0)$.
14. The paraboloid obtained by moving the surface $z = x^2 + y^2$ so that its vertex is at $(1, 3, 5)$, its axis is parallel to the x -axis, and the surface opens towards negative x values.

Problems

15. Consider the function f given by $f(x, y) = y^3 + xy$. Draw graphs of cross-sections with:

- (a) x fixed at $x = -1$, $x = 0$, and $x = 1$.
- (b) y fixed at $y = -1$, $y = 0$, and $y = 1$.

Problems 16–18 concern the concentration, C , in mg per liter, of a drug in the blood as a function of x , the amount, in mg, of the drug given and t , the time in hours since the injection. For $0 \leq x \leq 4$ and $t \geq 0$, we have $C = f(x, t) = te^{-t(5-x)}$.

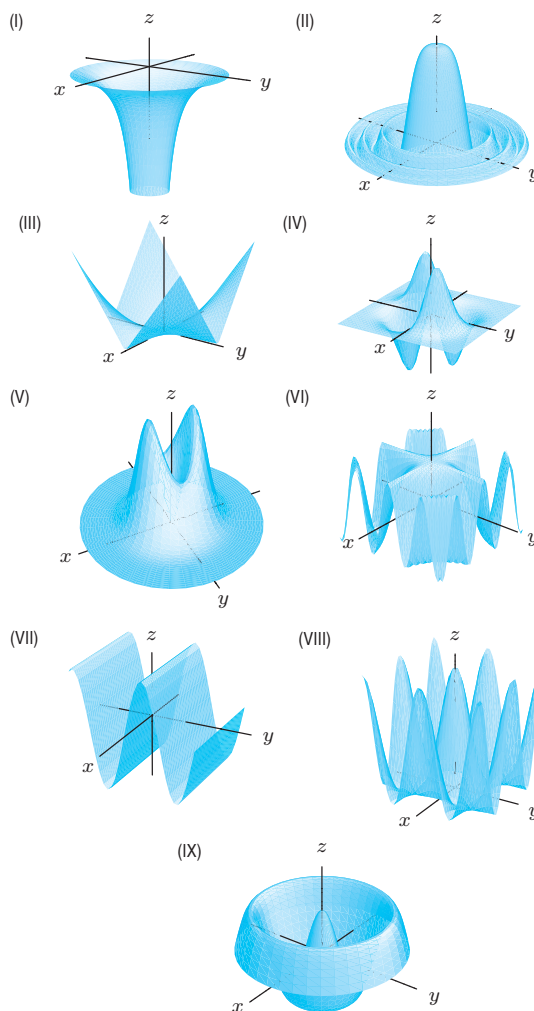
16. Find $f(3, 2)$. Give units and interpret in terms of drug concentration.
17. Graph the following single-variable functions and explain their significance in terms of drug concentration.

- (a) $f(4, t)$
- (b) $f(x, 1)$

18. Graph $f(a, t)$ for $a = 1, 2, 3, 4$ on the same axes. Describe how the graph changes as a increases and explain what this means in terms of drug concentration.

19. Without a computer or calculator, match the equations (a)–(i) with the graphs (I)–(IX).

- (a) $z = xye^{-(x^2+y^2)}$
- (b) $z = \cos(\sqrt{x^2+y^2})$
- (c) $z = \sin y$
- (d) $z = -\frac{1}{x^2+y^2}$
- (e) $z = \cos^2 x \cos^2 y$
- (f) $z = \frac{\sin(x^2+y^2)}{x^2+y^2}$
- (g) $z = \cos(xy)$
- (h) $z = |x||y|$
- (i) $z = (2x^2+y^2)e^{1-x^2-y^2}$



20. Decide whether the graph of each of the following equations is the shape of a bowl, a plate, or neither. Consider a plate to be any flat surface and a bowl to be anything that could hold water, assuming the positive z -axis is up.

(a) $z = x^2 + y^2$ (b) $z = 1 - x^2 - y^2$
 (c) $x + y + z = 1$ (d) $z = -\sqrt{5 - x^2 - y^2}$
 (e) $z = 3$

21. Sketch cross-sections for each function in Problem 20.

22. You like pizza and you like cola. Which of the graphs in Figure 12.30 represents your happiness as a function of how many pizzas and how much cola you have if

- (a) There is no such thing as too many pizzas and too much cola?
 (b) There is such a thing as too many pizzas or too much cola?
 (c) There is such a thing as too much cola but no such thing as too many pizzas?

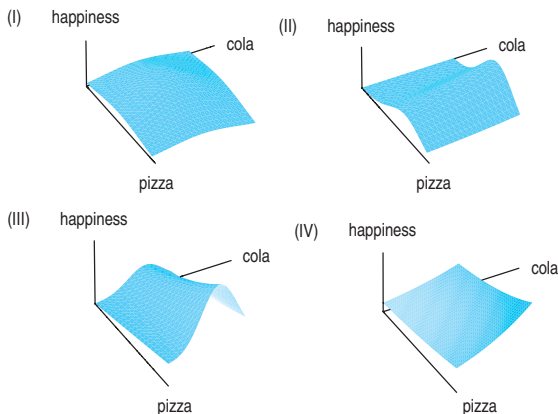


Figure 12.30

For Problems 23–26, give a formula for a function whose graph is described. Sketch it using a computer or calculator.

23. A bowl which opens upward and has its vertex at 5 on the z -axis.
 24. A plane which has its x -, y -, and z -intercepts all positive.
 25. A parabolic cylinder opening upward from along the line $y = x$ in the xy -plane.
 26. A cone of circular cross-section opening downward and with its vertex at the origin.

Strengthen Your Understanding

In Problems 32–33, explain what is wrong with the statement.

32. The graph of the function $f(x, y) = x^2 + y^2$ is a circle.
 33. Cross-sections of the function $f(x, y) = x^2$ with x fixed are parabolas.

27. Sketch cross-sections of $f(r, h) = \pi r^2 h$, first keeping h fixed, then keeping r fixed.

28. By setting one variable constant, find a plane that intersects the graph of $z = 4x^2 - y^2 + 1$ in a:

- (a) Parabola opening upward
 (b) Parabola opening downward
 (c) Pair of intersecting straight lines

29. For each of the graphs I–IV in Problem 22 draw:

- (a) Two cross-sections with pizza fixed
 (b) Two cross-sections with cola fixed.

30. A wave travels along a canal. Let x be the distance along the canal, t be the time, and z be the height of the water above the equilibrium level. The graph of z as a function of x and t is in Figure 12.31.

- (a) Draw the profile of the wave for $t = -1, 0, 1, 2$. (Put the x -axis to the right and the z -axis vertical.)
 (b) Is the wave traveling in the direction of increasing or decreasing x ?
 (c) Sketch a surface representing a wave traveling in the opposite direction.

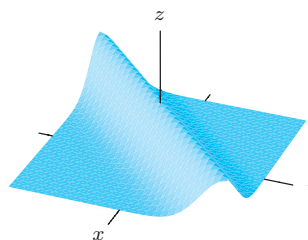


Figure 12.31

31. At time t , the displacement of a point on a vibrating guitar string stretched between $x = 0$ and $x = \pi$ is given by

$$f(x, t) = \cos t \sin x, \quad 0 \leq x \leq \pi, \quad 0 \leq t \leq 2\pi.$$

- (a) Sketch the cross-sections of this function with t fixed at $t = 0, \pi/4$ and the cross-sections with x fixed at $x = \pi/4, \pi/2$.
 (b) What is the value of f if $x = 0$ or $x = \pi$? Explain why this is to be expected.
 (c) Explain the relation of the cross-sections to the surface representing f .

In Problems 34–36, give an example of:

34. A function whose graph lies above the xy -plane and intersects the plane $z = 2$ in a single point.
 35. A function which intersects the xz -plane in a parabola and the yz -plane in a line.
 36. A function which intersects the xy -plane in a circle.

Are the statements in Problems 37–50 true or false? Give reasons for your answer.

37. The function given by the formula $f(v, w) = e^v/w$ is an increasing function of v when w is a nonzero constant.
38. A function $f(x, y)$ can be an increasing function of x with y held fixed, and be a decreasing function of y with x held fixed.
39. A function $f(x, y)$ can have the property that $g(x) = f(x, 5)$ is increasing, whereas $h(x) = f(x, 10)$ is decreasing.
40. The plane $x + 2y - 3z = 1$ passes through the origin.
41. The plane $x + y + z = 3$ intersects the x -axis when $x = 3$.
42. The sphere $x^2 + y^2 + z^2 = 10$ intersects the plane $x = 10$.
43. The cross-section of the function $f(x, y) = x + y^2$ with $y = 1$ is a line.
44. The function $g(x, y) = 1 - y^2$ has identical parabolas for all cross-sections with x constant.
45. The function $g(x, y) = 1 - y^2$ has lines for all cross-sections with y constant.
46. The graphs of $f(x, y) = \sin(xy)$ and $g(x, y) = \sin(xy) + 2$ never intersect.
47. The graphs of $f(x, y) = x^2 + y^2$ and $g(x, y) = 1 - x^2 - y^2$ intersect in a circle.
48. If all the cross-sections of the graph of $f(x, y)$ with x constant are lines, then the graph of f is a plane.
49. The only point of intersection of the graphs of $f(x, y)$ and $-f(x, y)$ is the origin.
50. The point $(0, 0, 10)$ is the highest point on the graph of the function $f(x, y) = 10 - x^2 - y^2$.
51. The object in 3-space described by $x = 2$ is

(a) A point	(b) A line
(c) A plane	(d) Undefined.

12.3 CONTOUR DIAGRAMS

The surface which represents a function of two variables often gives a good idea of the function's general behavior—for example, whether it is increasing or decreasing as one of the variables increases. However it is difficult to read numerical values off a surface and it can be hard to see all of the function's behavior from a surface. Thus, functions of two variables are often represented by contour diagrams like the weather map on page 666. Contour diagrams have the additional advantage that they can be extended to functions of three variables.

Topographical Maps

One of the most common examples of a contour diagram is a topographical map like that shown in Figure 12.32. It gives the elevation in the region and is a good way of getting an overall picture of the terrain: where the mountains are, where the flat areas are. Such topographical maps are frequently colored green at the lower elevations and brown, red, or white at the higher elevations.

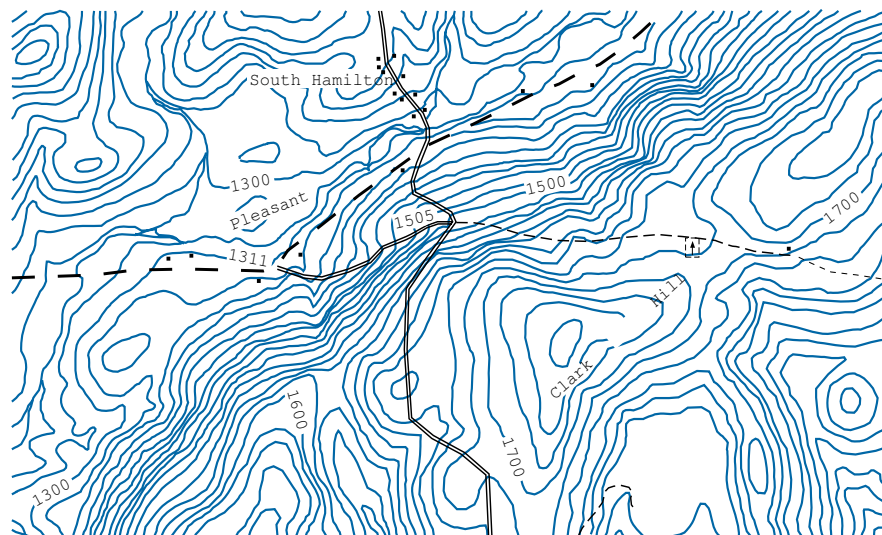


Figure 12.32: A topographical map showing the region around South Hamilton, NY